

Comparing 3D Data Representations for Keypoint Estimation in Humanoid Characters

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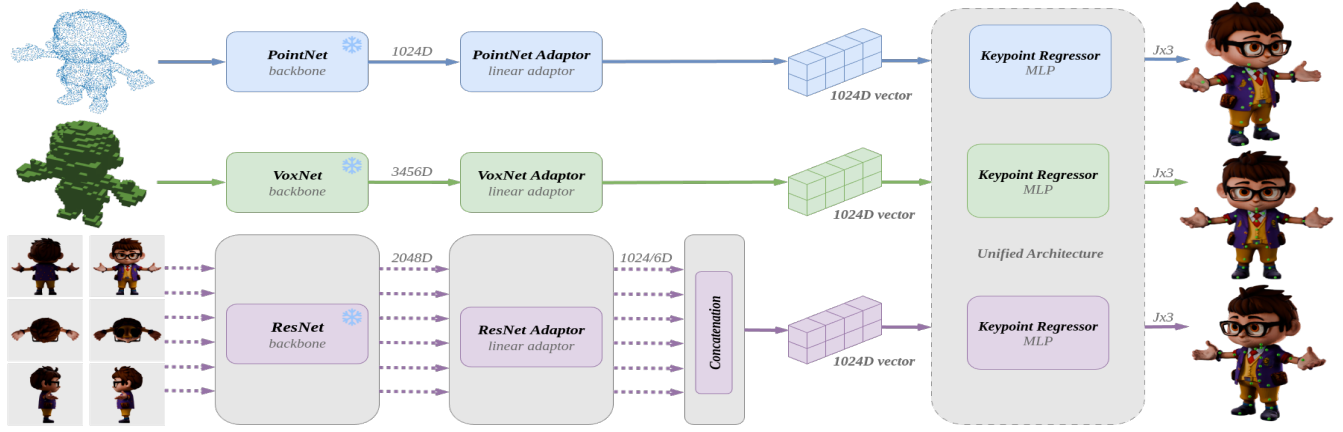


Figure 1: Overview of the keypoint prediction models' architectures. Each backbone is truncated to output latent vectors of the same size, and we train three separates heads (same architecture) to predict the 3D location of a standard set of keypoints.

ABSTRACT

This work investigates how the use of different 3D data representations affects the performance of learning-based models for the task of predicting keypoints from meshes of humanoid characters, used for rigging. From existing mesh datasets, we construct a multimodal dataset by extracting three modalities: multi-view images, voxel grids, and point clouds. For each modality, we train a neural network with a comparable amount of parameters, to find out which backbone best leverages the 3D information contained in the corresponding representation. Additionally, we estimate expert disagreement on the rigging task through a user study, which highlights the subjective nature of rigging and provides a practical upper bound to interpret model accuracy.

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1 INTRODUCTION

In recent years, neural networks have emerged as a solution for the task of estimating 3D keypoints from character meshes, automating the rigging process, which traditionally is a time-consuming manual process for human experts. While most approaches rig arbitrary characters [Xu et al. 2020, 2019; Zhang et al. 2025], some works focus on rigging humanoid characters [Chu et al. 2025; Kim et al. 2021; Ma and Zhang 2023], which have to strictly follow the same skeleton template. In both cases, neural networks rarely take meshes directly as an input: these methods use an intermediate 3D representation as input, whether voxels [Kim et al. 2021; Xu et al. 2019] or point-clouds [Chu et al. 2025; Xu et al. 2020; Zhang et al. 2025], or, even more recently, gaussian splats [Guo et al. 2025]. However, none of these approaches directly compare the different 3D input modalities available, nor do they evaluate how closely predictions can approximate the decisions made by expert artists.

In this work, we construct a multimodal dataset from existing datasets by extracting three geometric modalities: multi-view images, voxels, and point clouds, from a set of humanoid characters taken from HumanRig and RigNet. We train three separate networks, each specialized on one modality, using the same number of parameters for fair comparison. Our goal is to compare the amount

of information extracted from each modality by each backbone and conclude on which one perform better on the task at hand. As no implementation of HumanRig is currently released, we compare our networks to TARig [Ma and Zhang 2023], another state-of-the-art rigging method specific to humanoid characters. Also, to assess the limits of model performance, we conduct a rigging study, where several professional artists manually rig the same set of characters. This data reveals significant inter-artist variation, underscoring the subjectivity of the task and providing a basis to reinterpret standard evaluation metrics in light of human disagreement.

2 METHOD OVERVIEW

Our method predicts 3D skeletal keypoints from three geometric modalities. Every sample is processed by a modality-specific backbone that extracts specific geometric features. A linear adaptor compresses the backbone output into a 1024D vector for each modality, which is finally decoded by a multilayer perceptron into keypoints coordinates. All variants are trained end-to-end with an MSE loss on keypoints and evaluated with RigNet’s Chamfer distance metrics [Xu et al. 2020].

2.1 Multimodal dataset construction

Our multi-modal dataset is created with a pipeline that transforms rigged 3D character meshes from HumanRig [Chu et al. 2025] datasets into three target 3D representations. Each mesh is standardized into a canonical space, then converted into multiple formats:

- **Point clouds** are generated through Poisson disk sampling with area-weighted face selection, ensuring uniform surface coverage.
- **Voxels** use a 256^3 occupancy grid, employing multi-ray casting from each voxel center to robustly classify interior regions.
- **Multi-view images** capture six orthographic perspectives (front, back, left, right, top, bottom) rendered at 1080×1080 resolution.

The ground-truth positions of the bones is stored alongside with skinning information. Thanks to this pipeline, we created a dataset composed of 11,873 meshes from the RigNet (filtered to keep only humanoid characters, oftentimes manually corrected to fit the 22-joint human skeleton template) and HumanRig datasets. A sample of the dataset can be seen on the leftmost part of Fig. 1.

2.2 Keypoint regression networks

2.2.1 Multi view image network. Our image backbone is a ResNet-50 backbone pretrained on ImageNet, truncated before its classification head. The backbone is applied for each of the six RGB views, producing six 2048D descriptors that are linearly projected by a shared perceptron and concatenated to a single 1024D representation by our ResNet adaptor.

2.2.2 Voxel grid network. We process voxels using the VoxNet backbone [Maturana and Scherer 2015], which takes as input occupancy grids of shape 32^3 , and returns a 3456D feature vector, which is projected to 1024D by a perceptron ("VoxNet adaptor" on Fig. 1).

2.2.3 Point cloud network. We process point cloud using the PointNet backbone [Qi et al. 2017], which produces a 1024D feature vector, however we still apply a perceptron adaptor for fairness regarding the amount of trainable parameters.

For each modality, we train the adaptors along with a regression head (3-layer MLP) to predict the 3D position of each keypoint, totaling around 2M trainable parameters. Backbones are initialized with pretrained weights and kept frozen for evaluation fairness.

3 USER STUDY

RigNet proposed evaluation metrics based on the Chamfer distance, which have since been widely adopted to evaluate keypoint prediction models. However, previous works do not mention the impact of the subjective nature of rigging on those metrics : two artists might rig the same character in different ways. We conducted a user study amongst expert rigging artists, to estimate the natural variance in keypoint placement. For each metric, we quantify experts disagreement by computing the mean of the pairwise distances between human annotations on the same character, averaged over all 20 characters in our study.

We report this value in the first row of Table 1. This provides a practical lower bound on the set of metrics, caused by the variability in rigging preferences across human experts. Any model scoring below this lower bound would essentially be overfitting to a particular set of rigging preferences, which limits generalization.

Table 1: Evaluation metrics on our test set (1382 meshes)

Method	MAE↓	MSE↓	CD-J2J↓	CD-J2B↓	CD-B2B↓
Artists	0.0567	0.0265	0.0415	0.0348	0.0302
Ours (img)	0.0444	0.0013	0.0357	0.0262	0.0258
Ours (pc)	0.0532	0.0022	0.0418	0.0323	0.0315
Ours (vox)	0.0296	0.0007	0.0248	0.0166	0.0156
Ours (avg)	0.0360	0.0009	0.0298	0.0213	0.0208
RigNet	0.0302	0.0025	0.0302	0.0249	0.0245
TARig	0.0277	0.0020	0.0277	0.0222	0.02214



Figure 2: Predictions (voxel modality) on characters within and outside of the distribution of the training set.

4 RESULTS AND FUTURE WORK

Among the three input modalities we compared, the voxel-based model achieved the best overall performance, outperforming not only the point cloud and image models on every evaluation metric, but also the state-of-the-art TARig method, as reported in Table 1. This suggests that, out of the three modalities, the available pretrained backbones for voxels better encode the 3D information for the task of keypoint prediction. Finally, all model predictions fall below the human baseline for distance metrics, indicating that current models likely overfit to one specific rigging style. Future work could focus on adapting these networks to the artists’ preferred rigging conventions, as well as exploring additional input modalities, for instance graph-based backbones for meshes.

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